Revised Section 2.2 — Carbon Chains: Plastic Waste vs. Crude Oil

For: The Anthropogenic Reservoir White Paper

This section replaces the existing Section 2.2 in its entirety.

2.2 Carbon Chains: Plastic Waste vs. Crude Oil

The conventional pathway from crude oil to usable hydrocarbon product involves:

1. **Extraction** — drilling, pumping, artificial lift
2. **Transportation** — pipeline or tanker to refinery
3. **Distillation** — atmospheric and vacuum separation into fractions
4. **Cracking** — steam cracking of naphtha at 800°C+ to produce ethylene and propylene
5. **Polymerization** (if the end product is plastic) — catalytic assembly of monomers into polymer chains under heat and pressure

Each step carries energy costs, thermal losses, and infrastructure overhead. Approximately 4% of global oil and gas production is consumed as feedstock for plastics, and an additional 3–4% provides the energy for their manufacture.

Plastic waste to usable hydrocarbon chains via pyrolysis involves:

1. **Collection and sorting** — the primary logistical cost
2. **Thermal decomposition** — controlled heating at 400–600°C in the absence of oxygen

The pyrolysis step produces a hydrocarbon mixture — pyrolysis oil — with a heating value of 38–43 MJ/kg, comparable to petroleum diesel. This oil can feed directly into existing petrochemical infrastructure for further processing, or be used as fuel. Conversion rates of 75–85% from suitable plastic feedstock to liquid oil are achievable with current technology, with yields up to 85% demonstrated in fast pyrolysis processes.

The Energy Balance

The structural energy advantage of pyrolysis over the full crude-to-product chain is not a single ratio — it varies by end product, feedstock composition, and plant scale. However, the directional case is unambiguous.

Plastic waste retains a heating value of 41–47 MJ/kg — comparable to the 42–47 MJ/kg of crude oil from which it was originally derived. The carbon-hydrogen bonds are already formed. The molecular architecture is already partially organized. Pyrolysis does not need to overcome the thermodynamic barriers of geological extraction, refinery distillation, or initial molecular cracking. It operates on material that has already absorbed those energy investments.

The energy input for thermal decomposition of 1 kg of plastic is approximately 844 Wh (order-of-magnitude estimate; varies by reactor design and feedstock composition). The energy output — in the form of pyrolysis oil, syngas, and char — is approximately 9,500 Wh per kg under comparable conditions. The syngas fraction (a combustible gas byproduct of the pyrolysis reaction) can be recirculated to fuel the process itself, substantially reducing — and in optimized configurations approaching self-sustaining — external energy input for the thermal decomposition step.

This self-fueling characteristic is structurally significant. The crude oil pathway requires external energy at every stage — diesel for extraction equipment, fuel for tankers, heat for distillation columns, extreme temperatures for cracking. Pyrolysis, once initiated, partially sustains its own thermal requirements from byproducts of the reaction it is performing.

If the end product is fuel-grade oil: the energy savings are most pronounced. One thermal step replaces the entire extraction-to-refining sequence. The pyrolysis oil has properties compatible with conventional diesel for direct use in industrial furnaces, boilers, generators, and maritime engines, and can be blended with conventional diesel at meaningful ratios subject to fuel standard compliance and upgrading requirements.

If the end product is new polymer: the pyrolysis oil requires repolymerization — but the feedstock arrives at the polymerization step at lower total energy cost than virgin naphtha derived from crude oil. The oil can be refined using conventional petroleum refining processes and achieves quality comparable to virgin material, suitable for high-specification applications including food-grade packaging and pharmaceutical containers.

If the end product is chemical feedstock: pyrolysis oil serves as an input for producing new plastics and petrochemicals, closing the material loop. Major petrochemical firms — including Shell, ExxonMobil, Neste, and BASF — are actively building or acquiring pyrolysis capacity for precisely this purpose, with European plastics manufacturers planning to increase chemical recycling investment from €2.6 billion in 2025 to €8 billion by 2030.

The Sorting and Contamination Constraint

The primary constraint on plastic pyrolysis at scale is feedstock management — but this constraint is substantially less severe than commonly assumed.

Pyrolysis tolerates far higher contamination levels than mechanical recycling. Mechanical recycling requires clean, single-stream, source-separated plastic with minimal contamination — a requirement that excludes the vast majority of post-consumer plastic waste. Pyrolysis, by contrast, processes mixed and contaminated streams that mechanical recycling cannot handle, eliminating the need for washing and careful sorting that dominates mechanical recycling costs.

The sorting requirement for pyrolysis is not single-stream purity. It is exclusion of specific problematic polymer types. Low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), and polystyrene (PS) are well-suited pyrolysis feedstocks with high oil yields and clean decomposition. Polyvinyl chloride (PVC) produces hydrochloric acid during thermal decomposition, corroding equipment and contaminating the oil. Polyethylene terephthalate (PET) has unfavorable oxygen content that reduces oil quality.

The operational requirement is therefore a coarse sort — exclude PVC and PET, process the remainder — rather than the fine sort that mechanical recycling demands. This is a logistics optimization problem with active commercial solutions: automated optical sorting, density separation, and near-infrared spectroscopy systems are deployed at industrial scale and improving in throughput and accuracy. The problem is solvable with current engineering and is being solved — but it is not yet solved at the scale the feedstock inventory warrants. Honest acknowledgment of this gap is necessary: the technology exists, the deployment is scaling, and the trajectory is clear, but the claim that sorting is a fully resolved constraint at global scale would be premature.

Commercial Viability

Pyrolysis oil currently sells for $600–900 per ton. Syngas, the combustible gas byproduct, has a market value of $200–300 per ton and can generate up to 800 kWh of electricity per ton of waste processed. Carbon black, a solid byproduct, sells for $200–400 per ton. Actual revenue mix depends on plant configuration and product slate; not all facilities capture all value streams simultaneously. Against a feedstock cost of $30–200 per ton for waste plastic — or negative cost where tipping fees are available — the margin structure is favorable.

Feasibility studies under U.S. market conditions indicate that pyrolysis plants become profitable at scales above 60 tons per day. Profitability is sensitive to crude oil prices: at crude above $65 per barrel, pyrolysis-to-fuel is commercially viable without green premiums or policy incentives. Below that threshold, tipping fees, carbon credits, or regulatory support improve the business case but are not structurally required at the energy-economics level.

The pyrolysis equipment market is growing at approximately 27% annually, with over $4.1 billion invested in pyrolysis infrastructure in 2023. Equipment prices declined 12% between 2021 and 2024 due to increased competition and local manufacturing. Modular plant

designs — enabling scalable deployment matched to local feedstock availability — saw 47% adoption growth. Over 430 mobile pyrolysis units have been commissioned globally, and more than 50 joint ventures between pyrolysis technology firms and waste collection companies were formed in 2023 alone.

This is not a nascent technology awaiting breakthrough. It is a proven process undergoing rapid commercial scale-up, driven by the same energy-cost arbitrage this paper describes: the anthropogenic reservoir of plastic waste is a hydrocarbon source with known composition, zero exploration risk, growing inventory, and structural cost advantages over geological extraction that widen as oil reserves become more expensive to access.

Revised Unified Gradient Table Entry

Material	Primary Energy (Geological)	Secondary Energy (Anthropogenic)	Approximate Savings
Plastic (to fuel-grade oil)	Full extraction + transportation + distillation + cracking chain	Pyrolysis at 400–600°C, partially self-fueled via syngas recovery	Structural: bypasses 4 of 5 primary processing steps; energy input ~844 Wh/kg vs. energy output ~9,500 Wh/kg; net savings plant- and feedstock-dependent but directionally large
Plastic (to new polymer)	Full extraction + refining + cracking + polymerization chain	Pyrolysis + repolymerization	Feedstock arrives at polymerization at lower total energy cost than virgin naphtha; savings reduced by repolymerization step but remain structurally positive

Note: This revised section replaces the "~50–70% (estimated)" figure in the original table with a structural description of the energy advantage. The advantage is real and directionally large, but a single percentage obscures the dependence on end product, feedstock composition, and plant configuration. The structural description is more defensible and more informative for the target audience.